Bioinf Pipelines. HW1

Performed by Victor Florin

1. Install nextflow

Checking for the necessary components:

- [x] ~~Conda~~

```
(base) student@LAPTOP-NKRTACMG:~$ conda --version
conda 26.1.1
(base) student@LAPTOP-NKRTACMG:~$
```

- [x] ~~Java & Nextflow~~

```
student@LAPTOP-NKRTACMG:~$ nextflow info
  Version: 25.10.4 build 11173
  Created: 10-02-2026 15:17 UTC (18:17 MSD)
  System: Linux 5.10.16.3-microsoft-standard-WSL2
  Runtime: Groovy 4.0.28 on OpenJDK 64-Bit Server VM 17.0.18+8-Ubuntu-124.04.1
  Encoding: UTF-8 (UTF-8)
```

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2. Check out rnaseq-nf pipeline

Link: https://www.nextflow.io/docs/latest/tutorials/rnaseq-nf.html

Cloning the repository with the pipeline:

```
student@LAPTOP-NKRTACMG:~$ cd ~
student@LAPTOP-NKRTACMG:~$ git clone https://github.com/nextflow-io/rnaseq-nf.git
Cloning into 'rnaseq-nf'...
remote: Enumerating objects: 1030, done.
remote: Counting objects: 100% (630/630), done.
remote: Compressing objects: 100% (237/237), done.
remote: Total 1030 (delta 411), reused 533 (delta 341), pack-reused 400 (from 2)
Receiving objects: 100% (1030/1030), 664.49 KiB | 4.34 MiB/s, done.
Resolving deltas: 100% (502/502), done.
student@LAPTOP-NKRTACMG:~$ cd rnaseq-nf
```

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3. Run the pipeline on the test data in:

- Docker;
- Conda environment.

Checking the Docker version:

```
C:\Users\Student>docker --version
Docker version 29.3.1, build c2be9cc
```

- [x] ~~Conda:~~

```
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ nextflow run main.nf -profile conda --outdir results_conda

N E X T F L O W  ~  version 25.10.4

Launching `main.nf` [nasty_easley] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/student/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/student/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results_conda

[-        ] RNASEQ:INDEX  -
[-        ] RNASEQ:FASTQC -
executor >  local (3)
executor >  local (4)
[b4/02c5de] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✓
[2d/c49e6b] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✓
[66/c658f2] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✓
[e6/aedce3] MULTIQC                                 | 1 of 1 ✓

Done! Open the following report in your browser --> results_conda/multiqc_report.html

Completed at: 09-Apr-2026 18:36:28
Duration    : 3m 47s
CPU hours   : (a few seconds)
Succeeded   : 4


(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ ls -la results_conda/
total 4976
drwxr-xr-x  3 student student    4096 Apr  9 18:36 .
drwxr-xr-x 12 student student    4096 Apr  9 18:36 ..
drwxr-xr-x  2 student student    4096 Apr  9 18:34 fastqc_ggal_gut_logs
-rw-r--r--  1 student student 5079861 Apr  9 18:36 multiqc_report.html
```

~~Success. I will add the contents of the~~ `~~\results_conda~~` ~~folder to GitHub.~~

- [x] ~~Docker:~~

```
(base) student@LAPTOP-NKRTACMG:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(base) student@LAPTOP-NKRTACMG:~$ cd ~/rnaseq-nf
nextflow(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ nextflow run nextflow-io/rnaseq-nf -r master -profile docker --outdir results_docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [peaceful_faraday] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/student/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/student/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results_docker

executor >  local (4)
[dd/d281b8] RNA…INDEX (ggal_1_48850000_49020000) | 1 of 1 ✓
[66/afaf02] RNASEQ:FASTQC (ggal_gut)             | 1 of 1 ✓
[f1/9360c2] RNASEQ:QUANT (ggal_gut)              | 1 of 1 ✓
[33/dde3bc] MULTIQC                              | 1 of 1 ✓

Done! Open the following report in your browser --> results_docker/multiqc_report.htm

Completed at: 09-Apr-2026 19:29:02
Duration    : 3m
CPU hours   : 0.1
Succeeded   : 4
```

~~Success. I will add the contents of the~~ `~~\results_docker~~` ~~folder to GitHub.~~

Observation:

Conda: 3m 47s

Docker: 3m 0s

Docker completed faster.

4. Download any RNA-seq dataset from NCBI (but not to big) and run the pipeline with any profile on it

I selected **GSE156072** (Non-Canonical Epigenetic Regulation of HIV Infection in Macrophage, Homo sapiens):

- 12 samples in total;
- 1 control group (uninfected CD4 T cells);
- 1 experimental group (HIV infected CD4 T cells);
- 3 replicates.

For the sake of speed, I decided to consider only the **SRR12432181** sample (**SRR12432181_1.fastq + SRR12432181_2.fastq**).

The pipeline was launched with the Conda profile, as this profile demonstrated stable performance on the test data. Docker is slightly faster, but it does occasionally encounter issues.

When attempting to run Docker Desktop on Windows 11, a known issue arose: the application would hang at the "Starting engine" stage and fail to initialize WSL2 components. After analyzing the logs and documentation, we decided to install Docker Engine directly on the Ubuntu WSL2 distribution. This circumvented issues with the graphical interface and network integration, ensuring stable operation of Docker for pipeline execution.

Additionally, we had to run the Docker daemon with the `--iptables=false` flag due to incomplete iptables support in the WSL2 kernel. After these settings, the pipeline ran successfully in the Docker environment.

Initially, **the pipeline used the chicken test transcriptome** and test reads by default, rather than my data from GSE156072.

```
(base) student@LAPTOP-NKRTACMG:~/rnaseq_data$ cd ~/rnaseq-nf
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ nano samplesheet.csv
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ cat samplesheet.csv
sample,fastq_1,fastq_2
GSE156072_sample,/home/student/rnaseq_data/SRR12432181_1.fastq.gz,/home/student/rnaseq_data/SRR12432181_2.fastq.gz
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$ nextflow run main.nf -profile conda --input samplesheet.csv --outdir results_my_data

 N E X T F L O W  ~  version 25.10.4

Launching `main.nf` [soggy_khorana] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/student/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/student/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results_my_data

executor >  local (4)
[a5/d7ed54] RNA…INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[e2/55ba7a] RNASEQ:FASTQC (ggal_gut)             | 1 of 1 ✔
[36/4f5c01] RNASEQ:QUANT (ggal_gut)              | 1 of 1 ✔
[f0/36677a] MULTIQC                              | 1 of 1 ✔

Done! Open the following report in your browser --> results_my_data/multiqc_report.html
```

To fix this, **I downloaded the human transcriptome and then rewrote the commands to account for this**.

```
gunzip Homo_sapiens.GRCh38.cdna.all.fa.gz--2026-04-09 20:33:20--  ftp://ftp.ensembl.org/pub/release-113/fasta/homo_sapiens/cdna/Homo_sapiens.GRCh38.cdna.all.fa.gz
           => 'Homo_sapiens.GRCh38.cdna.all.fa.gz'
Resolving ftp.ensembl.org (ftp.ensembl.org)... 193.62.193.169
Connecting to ftp.ensembl.org (ftp.ensembl.org)|193.62.193.169|:21... connected.
Logging in as anonymous ... Logged in!
==> SYST ... done.    ==> PWD ... done.
==> TYPE I ... done.  ==> CWD (1) /pub/release-113/fasta/homo_sapiens/cdna ... done.
==> SIZE Homo_sapiens.GRCh38.cdna.all.fa.gz ... 79685174
==> PASV ... done.    ==> RETR Homo_sapiens.GRCh38.cdna.all.fa.gz ... done.
Length: 79685174 (76M) (unauthoritative)

Homo_sapiens.GRCh38.c 100%[=====================>]  75.99M   641KB/s    in 59s

executor >  local (4)
executor >  local (4)
[9a/a27a89] RNASEQ:INDEX (Homo_sapiens) | 1 of 1 ✔
[08/16441b] RNASEQ:FASTQC (SRR12432181) | 1 of 1 ✔
[42/9dc974] RNASEQ:QUANT (SRR12432181)  | 1 of 1 ✔
[e6/cf18cf] MULTIQC                     | 1 of 1 ✔

Done! Open the following report in your browser --> results_my_data_human/multiqc_report.html

Completed at: 09-Apr-2026 21:15:32
Duration    : 40m 32s version 25.10.4
CPU hours   : 0.8
Succeeded   : 4.nf` [hopeful_watson] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$
transcriptome: /home/student/rnaseq-nf/Homo_sapiens.GRCh38.cdna.all.fa
reads        : /home/student/rnaseq_data/SRR12432181_{1,2}.fastq.gz
outdir       : results_my_data_human

executor >  local (2)
[9a/a27a89] RNASEQ:INDEX (Homo_sapiens) | 0 of 1
[08/16441b] RNASEQ:FASTQC (SRR12432181) | 0 of 1
[          ] RNASEQ:QUANT               -
(base) student@LAPTOP-NKRTACMG:~/rnaseq-nf$
```

The report is saved in the folder `~/rnaseq-nf/results_my_data_human/`.

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5. Prepare a report with screenshots of the work that has been done (PDF, Markdown, Word)

Done - Obsidian note.

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6. Create new GitHub repo for this course

Done.

chaosintoorder/Bioinformatics-Pipelines

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7. Upload report to hw1 folder in this repo

Done.